2.3 Camera Calibration Self-Check Quiz

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Submitted Nov 13, 7:35 PM IST **Attempts** Unlimited

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1. What is camera calibration in computer vision? **1 / 1 point**

- ○ The process of adjusting the focus of a camera lens
- ◉ The process of determining the projection matrix
- ○ The process of converting image coordinates to world coordinates
- ○ The process of resizing images to a specific resolution

Correct
Camera calibration in computer vision refers to the process of determining the intrinsic and extrinsic parameters of a camera. These parameters are essential for accurately mapping 3D points in the real world to 2D points in an image and vice versa.

2. When performing camera calibration, we set up a system of equations $A||\mathbf{p}|| = \mathbf{0}$ in the parameters $\mathbf{p}$ that define the camera projection matrix. We then try to minimize $A||\mathbf{p}||$ subject to $||\mathbf{p}|| = 1$. Here, we constrain $||\mathbf{p}|| = 1$ because: **1 / 1 point**

- ○ A camera projection matrix is valid only if its Frobenius norm is 1.
- ○ The constraint makes the optimization easier to implement.
- ○ The correspondences used to form A might be noisy.
- ◉ The equations $A||\mathbf{p}|| = \mathbf{0}$ are not sufficient to produce a unique matrix P, and will produce a family of solutions.

Correct
Because the projection equations are all expressed in homogeneous coordinates, if P is a solution, so is αP. To isolate a single solution, we need to add a constraint on the Frobenius norm.